EXPERIMENT-09: Hiding and Extracting a Text File Behind an Image/Audio File Using Command Prompt

AIM:

To study how a text file can be appended to an image or audio file using Windows Command Prompt and how the appended data can be extracted for forensic analysis.

TOOLS REQUIRED:

1. Windows Command Prompt (cmd)
2. Sample text file (A.txt)
3. Sample image (B.jpg)
4. Optional audio file (sound.mp3) and archive (secret.zip)
5. Notepad and an archive utility

PROCEDURE:

Step 1: Prepare the Files

1. Create a text file named A.txt containing sample text.
2. Copy an image named B.jpg to the same folder.
3. For the optional audio experiment, prepare sound.mp3 and secret.zip.

Step 2: Open Command Prompt

1. Press **Win + R**.
2. Type cmd and press Enter.
3. Navigate to the Desktop:
4. cd %USERPROFILE%\Desktop

Step 3: Append the Text to an Image

1. Execute:
2. copy /b B.jpg + A.txt C.jpg
3. This creates C.jpg by appending the contents of A.txt to the end of B.jpg. The visible image may still open normally in many image viewers.

Step 4: Inspect the Appended Text

1. Locate C.jpg.
2. Open a copy of the file in Notepad.
3. Scroll toward the end of the file.
4. Check for the appended text from A.txt.

Step 5: Append a File to Audio (Optional)

1. Execute: copy /b sound.mp3 + secret.zip newfile.mp3
2. This appends the ZIP data to the audio file. Compatibility varies by media player and archive utility.

Step 6: Verify the Experiment

1. Confirm that the output files were created.
2. Check whether the original image or audio remains playable.
3. Verify that the appended data is present.
4. Record the observations and capture screenshots.

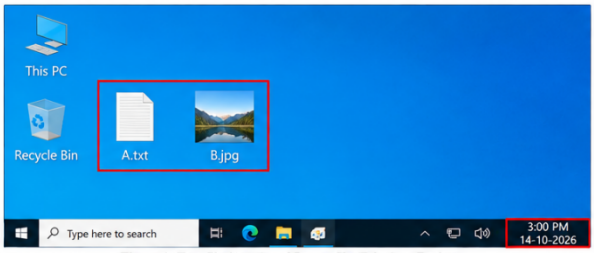


Figure 1: Text file (A.txt) and Image file (B.jpg) on Desktop

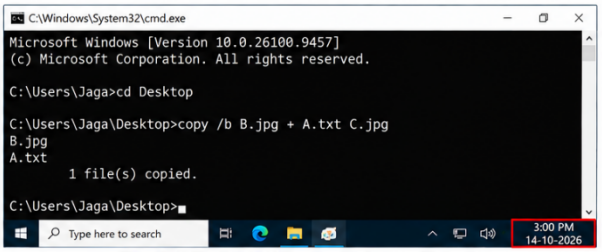


Figure 2: Hiding Text inside Image using Command Prompt

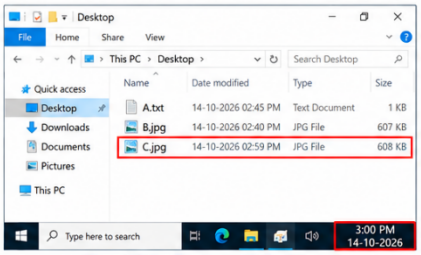


Figure 3: Merged Image File (C.jpg) Created

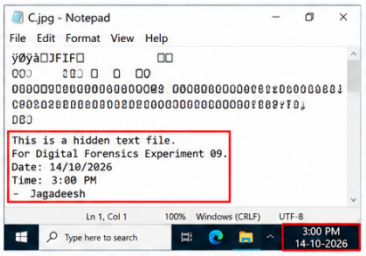


Figure 4: Viewing Hidden Text inside C.jpg (Notepad)

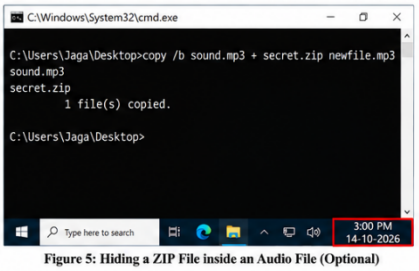


Figure 5: Hiding a ZIP File inside an Audio File (Optional)

RESULT:

1. The text file was successfully hidden within the image file without altering the visible content.
2. The hidden data could be retrieved by opening the merged file in a text editor.
3. This basic stegnography technique demonstrates how files can be concealed in multimedia content for confidentiality or forensic challenges.